

nAD10205-13a

10-bit 205 MSPS Analog-to-Digital Converter IP

FEATURES

- TSMC CL013G 1.2 V Technology
- 10-bit Pipeline ADC
- 25 to 205 MSPS Conversion Rate
- Excellent Dynamic Performance
58 dBFS SNR at $F_{IN} = 10$ MHz
68 dBc SFDR at $F_{IN} = 10$ MHz
- Dynamic Power Scaling
~0.35 mW per MSPS
- Low Power Consumption
72 mW at 205 MSPS
28 mW at 80 MSPS
- Power Saving Idle Modes
- 0.5 mm² Core Area
- Pure Internal Voltage Reference
- Programmable Input Swing
0.5 to 1 V p-p Differential
- Input Clock Duty Cycle Restorer

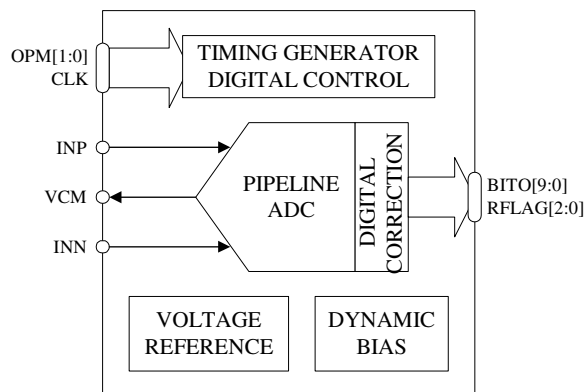


Figure 1. Functional block diagram

APPLICATIONS

- RGB / YUV High Speed Graphic Capture
Up to UXGA Resolution
- Communication Receive Channel
WLAN 802.11x / WiMAX 802.16x
- TV / Video / Radio Decoders

GENERAL DESCRIPTION

The nAD10205-13a is a monolithic, high-speed, ultra low power, analog-to-digital converter silicon IP. It uses a fully differential multistage pipeline architecture with digital error correction. The core includes a sample-and-hold and an internal voltage reference that provides a nominal full-scale range of 1.0 V peak-to-peak.

The ADC is designed for high dynamic performance at input frequencies up to Nyquist. It thus represents an ideal solution for demanding applications like RGB / YUV high speed

graphic capture, broadband communication and TV/Video decoders.

The ADC consumes only 72 mW at 205 MSPS operation. Dynamic power scaling means that the power consumption scales linearly with approximately ~0.35 mW per MSPS. Combined with power saving idle modes the ADC is suitable for battery powered devices. Output data is available in a binary offset coded format. Three out-of-range indicator bits are also available for determining if the input signal is over-range, under-range or out-of-range.

Implemented in a generic 0.13 μ m CMOS process, employing a fully differential architecture and no required external reference decoupling it represents an ideal ADC for highly integrated mixed-signal systems.

QUICK REFERENCE DATA

IP Type / Technology	Hard Macro / TSMC CL013G 1.2 V 1P6M			
IP Area / Dimensions	0.5 mm ² / 1.2 × 0.4 mm			
Parameter	Min.	Typ.	Max.	Unit
Power Dissipation, at 205 MSPS		72		mW
Signal-to-Noise Ratio, at 205 MSPS $F_{IN} = 10$ MHz		58		dBFS
Spurious-Free-Dynamic Range, 205 MSPS $F_{IN} = 10$ MHz		68		dBc
Effective Number of bits, 205 MSPS $F_{IN} = 10$ MHz		9.3		Bit

Table 1. nAD10205-13a quick reference data

**ABOUT NORDIC SEMICONDUCTOR**

Founded in 1983, Nordic Semiconductor ASA (OSE:NOD) is a Fabless Semiconductor specializing in short-range radio communication and high-speed data conversion. Nordic is head quartered in Trondheim, Norway,

MIXED SIGNAL IP OFFERINGS

Leading and emerging Fabless Semiconductor companies around the globe leverage Nordic's portfolio of off-the-shelf mixed signal intellectual property for the design of highly integrated circuits for broadband communication, digital imaging and video.

Our IP portfolio includes medium resolution (8 to 12-bit), high-speed (40 MSPS+) data converters (ADC and DAC) and low jitter clock generation (PLL) implemented in the latest deep submicron CMOS technologies. Featuring low power consumption, low-silicon area, and a minimum of required external components and technology options our products offers cost efficient integration of high-performance data converters in cost sensitive, high volume integrated circuit designs.

DELIVERABLES

Our mixed signal intellectual property is delivered as technology specific hard macros. Upon licensing we provide a complete design-kit and documentation that enables efficient, low risk, design-in and integration. The design kit for our off-the-shelf IPs includes:

- Full datasheet
- Evaluation board and samples
- Silicon validation report
- Integration Guidelines
- Physical design database (flat gds2)
- LVS Netlist (spice compatible)
- Footprint (.lef format)
- HDL model (verilog model)
- Timing model (.lib)
- Design-in and integration support

FURTHER INFORMATION

For further information about this product, such as full datasheet, silicon validation report, availability and licensing terms please contact us at: datacon@nordicsemi.no

DOCUMENT INFO	
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